Assignment 8: Probability Concepts

Load the Dataset and Answer the Following Questions:

1. How many null values are there in each column?

    1.1. Replace the null value of column "wheelbase" with the max value of the same column.

    1.2. Replace the null value of column "carwidth" with the Min value of the same column.

    1.3. Use Linear Interpolation to change the null value of column "carheight".

    1.4. Replace the null value of column "enginesize" with the mod of the same column.

2. What is the Probability that a Car is Sedan ?

3. What is the Probability that a car is Sedan and works with gas ?

4. What is the probability that a car is Sedan given it is a gas fuel type?

5. Given that the car height is higer than 65, find the probability that the car engin size is greater than 130?

Part 1: Handling missing values & Basic concepts in Probability

Import necessary libraries, load the 'Cars.csv' dataset, display a data sample, determine dataset dimensions, and determine the summary statistics for quantitative and qualtitative columns.

In [116…

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
import warnings
warnings.filterwarnings("ignore")
```

In [117…

```
df = pd.read_csv('Cars.csv')
print(df.shape)
df.head()
```

```
(205, 24)
```

Out[117…

	carname	fueltype	aspiration	doornumber	carbody	drivewheel	enginelocation	wheelbase	carlength	carwidth	...	eng
0	alfa-romero giulia	gas	std	two	convertible	rwd	front	88.6	168.8	64.1	...	
1	alfa-romero stelvio	gas	std	two	convertible	rwd	front	88.6	168.8	64.1	...	
2	alfa-romero Quadrifoglio	gas	std	two	hatchback	rwd	front	94.5	171.2	65.5	...	
3	audi 100 ls	gas	std	four	sedan	fwd	front	99.8	176.6	66.2	...	
4	audi 100ls	gas	std	four	sedan	4wd	front	99.4	176.6	66.4	...	

5 rows × 24 columns

In [118…

```
df.describe()
```

Out [118...

	wheelbase	carlength	carwidth	carheight	curbweight	enginesize	boreratio	stroke	compressionratio	h
count	194.000000	205.000000	185.000000	190.000000	205.000000	185.000000	205.000000	205.000000	205.000000	2
mean	98.669588	174.049268	65.932973	53.738947	2555.565854	124.081081	3.329756	3.255415	10.142537	1
std	5.934168	12.337289	2.113143	2.470452	520.680204	38.528771	0.270844	0.313597	3.972040	
min	86.600000	141.100000	60.300000	47.800000	1488.000000	61.000000	2.540000	2.070000	7.000000	4
25%	94.500000	166.300000	64.200000	52.000000	2145.000000	97.000000	3.150000	3.110000	8.600000	1
50%	97.000000	173.200000	65.500000	54.100000	2414.000000	110.000000	3.310000	3.290000	9.000000	9
75%	101.800000	183.100000	66.600000	55.575000	2935.000000	141.000000	3.580000	3.410000	9.400000	1
max	120.900000	208.100000	72.300000	59.800000	4066.000000	308.000000	3.940000	4.170000	23.000000	2

In [119...

```
df.describe(include=['object'])
```

Out [119...

	carname	fueltype	aspiration	doornumber	carbody	drivewheel	enginelocation	enginetype	cylindernumber	fuelsystem
count	205	205	205	205	205	205	205	205	205	205
unique	147	2	2	2	5	3	2	7	7	8
top	toyota corona	gas	std	four	sedan	fwd	front	ohc	four	mpfi
freq	6	185	168	115	96	120	202	148	159	94

Q1: How many null values are there in each column?

In [120...

```
df.isna().sum()
```

Out[120…
```
carname              0
fueltype             0
aspiration           0
doornumber           0
carbody              0
drivewheel           0
enginelocation       0
wheelbase           11
carlength            0
carwidth            20
carheight           15
curbweight           0
enginetype           0
cylindernumber       0
enginesize          20
fuelsystem           0
boreratio            0
stroke               0
compressionratio     0
horsepower           0
peakrpm              0
citympg              0
highwaympg           0
price                0
dtype: int64
```

In [121…
```
df['wheelbase'] = df['wheelbase'].fillna(df['wheelbase'].max())
```

In [122…
```
df['carwidth'] = df['carwidth'].fillna(df['carwidth'].min())
```

In [123…
```
df['carheight'] = df['carheight'].interpolate(method='linear')
```

In [124…
```
df['enginesize'] = df['enginesize'].fillna(df['enginesize'].mode()[0])
df.isna().sum()
```

Out[124…
```
carname             0
fueltype            0
aspiration          0
doornumber          0
carbody             0
drivewheel          0
enginelocation      0
wheelbase           0
carlength           0
carwidth            0
carheight           0
curbweight          0
enginetype          0
cylindernumber      0
enginesize          0
fuelsystem          0
boreratio           0
stroke              0
compressionratio    0
horsepower          0
peakrpm             0
citympg             0
highwaympg          0
price               0
dtype: int64
```

Q2: What is the Probability that the Car is Sedan ? (1 mark)

In [125…
```python
Pa = (df.carbody == 'sedan').mean();
Pa
```

Out[125…
```
np.float64(0.4682926829268293)
```

Q3: What is the Probability that the car is Sedan and works with gas?

In [126…
```python
Pb = (df.fueltype == 'gas').mean()
print(Pa * Pb )
```

```
0.4226055919095777
```

Q4:What is the probability that a car is Sedan given it is a gas fuel type?

In [127…
```
(Pa * Pb) / Pb
```

Out[127…
```
np.float64(0.46829268292682935)
```

Q5: Given that the car height is higer than 50, find the probability that the car engine size is greater than 130?

In [128…
```
# P(sz > 130 | height > 50)
Pa = (df.enginesize > 130).mean()
Pb = (df.carheight > 50).mean()
(Pa * Pb) / Pb
```

Out[128…
```
np.float64(0.28780487804878047)
```

Part 2: Probability Functions

Q6: Load dataset 'students.csv' and show first rows, shape, and compute the skewness.

In [129…
```
df2 = pd.read_csv('students.csv')
df2.shape
```

Out[129…
```
(8239, 16)
```

In [130…
```
df2.head()
```

Out[130…

	stud.id	name	gender	age	height	weight	religion	nc.score	semester	major	minor	score1	score2	on
0	833917	Gonzales, Christina	Female	19	160	64.8	Muslim	1.91	1st	Political Science	Social Sciences	NaN	NaN	
1	898539	Lozano, T'Hani	Female	19	172	73.0	Other	1.56	2nd	Social Sciences	Mathematics and Statistics	NaN	NaN	
2	379678	Williams, Hanh	Female	22	168	70.6	Protestant	1.24	3rd	Social Sciences	Mathematics and Statistics	45.0	46.0	
3	807564	Nem, Denzel	Male	19	183	79.7	Other	1.37	2nd	Environmental Sciences	Mathematics and Statistics	NaN	NaN	
4	383291	Powell, Heather	Female	21	175	71.4	Catholic	1.46	1st	Environmental Sciences	Mathematics and Statistics	NaN	NaN	

In [131…

```
skewness = df2.skew(numeric_only=True)
print(skewness.sort_values())
```

```
score1            -0.318819
score2            -0.304878
height            -0.009244
stud.id            0.018612
salary             0.273835
nc.score           0.446834
online.tutorial    0.467502
weight             0.651004
graduated          1.403900
age                4.427823
dtype: float64
```

In [132…

```
df2.isna().sum()
```

Out[132…
```
stud.id                0
name                   0
gender                 0
age                    0
height                 0
weight                 0
religion               0
nc.score               0
semester               0
major                  0
minor                  0
score1              3347
score2              3347
online.tutorial        0
graduated              0
salary              6486
dtype: int64
```

Q7: What is the probability that the salary is greater than 20000 and less than 30000 ?

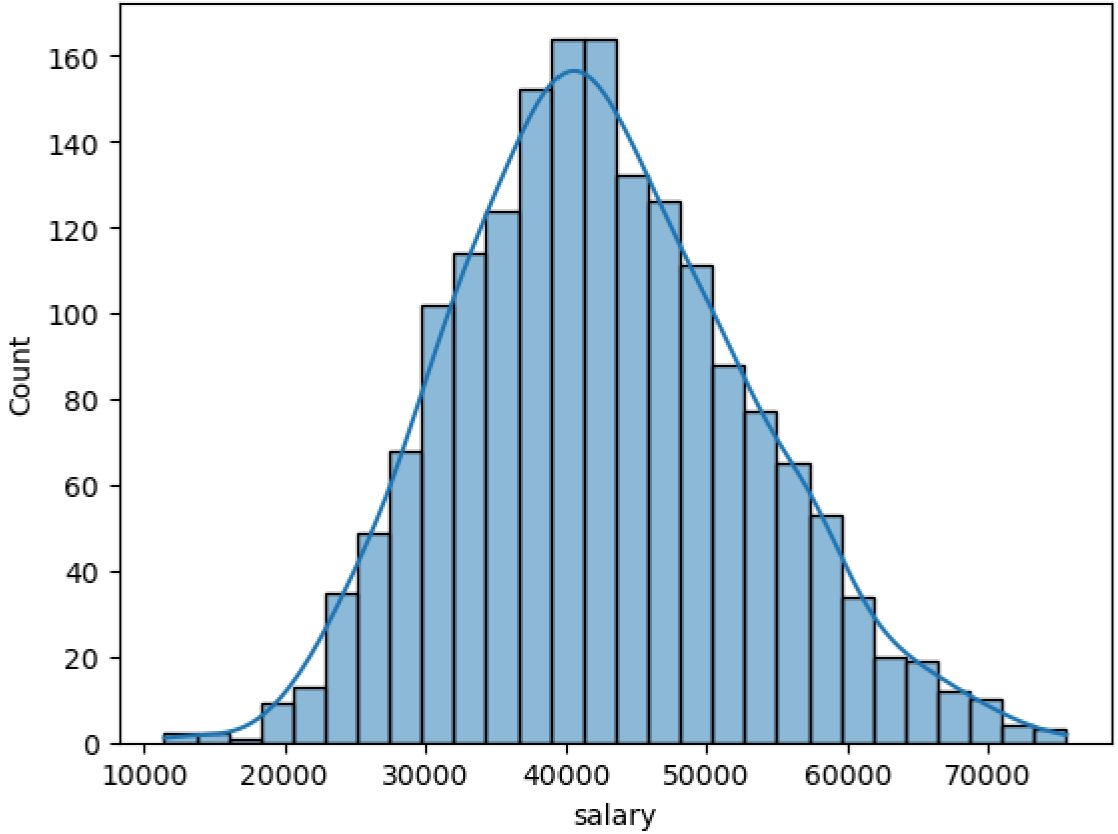
To answer this question, you should do the following:

1. Identify the distribution by
    - Plotting Histogram
    - calculate the skewness
    - Which distribution does this data follow?
2. Compute mean and std
3. Convert to Z-scores
4. Use CDF difference with the use of the right distribution function from scipy.stats

1. Distribution

In [133…
```
sns.histplot(df2, x='salary', kde=True)
```

Out[133…
```
<Axes: xlabel='salary', ylabel='Count'>
```



In [134…
```
salary_skew = df2['salary'].dropna().skew()
print(salary_skew)
```

```
0.2738351416144119
```

Since the -0.5 < salary_skew < 0.5, it is normally distributed

2. mean, std

In [135…
```
print(df2['salary'].mean())
print(df2['salary'].std())
```

```
42522.112364181405
10333.139905547336
```

3. Converting to Z score

In [136…

```
mu = df2['salary'].mean()
sigma = df2['salary'].std()

df2['salary_z'] = (df2['salary'] - mu) / sigma
df2[['salary', 'salary_z']].head()
```

Out[136…

	salary	salary_z
0	NaN	NaN
1	NaN	NaN
2	NaN	NaN
3	NaN	NaN
4	NaN	NaN

4. CDF difference with the use of the right distribution function from scipy.stats

In [137…

```
from scipy.stats import norm

lower_bound = 20000
upper_bound = 30000

z1 = (lower_bound - mu) / sigma
z2 = (upper_bound - mu) / sigma
p = norm.cdf(z2) - norm.cdf(z1)
p
```

Out[137…

```
np.float64(0.09814325462173662)
```

P(20000< salary < 30000) = 9.8%

Q8: A factory produces 200 chips with defect rate 8%. Answer the following questions:

1. What is the distribution name of this problem, based on the following questions?
2. What is the probability of getting exactly 14 defective chips?
3. At least 14 defective?
4. At most 15 defective?
5. Expected defects and standard deviation

You should use the right distribution function from scipy.stats to proceed with the answers

1- The distribution name of this problem is `Binomial`

In [138…

```
from scipy.stats import binom
import math
```

P(x==14)

In [139…

```
binom.pmf(14, 200, 0.08)
```

Out[139…

```
np.float64(0.09541156492261971)
```

P(x>=14)

In [140…

```
1 - binom.cdf(13, 200, 0.08)
```

Out[140…

```
np.float64(0.7357078859493741)
```

P(x<=15)

In [141…

```
binom.cdf(15, 200, 0.08)
```

Out[141… np.float64(0.4625822359332888)

Expected defects and standard deviation

In [142…
```
print("Expected defects: ", 200*0.08)
print("Std: ", math.sqrt(200*0.08*0.92))
```

```
Expected defects:  16.0
Std:  3.8366652186501757
```